6. Write a C program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U.”Break this code.

Aim:

To break an affine cipher using the most frequent and second most frequent ciphertext letters.

Procedure:

1. Assume the most frequent ciphertext letter B represents E.
2. Assume the second most frequent ciphertext letter U represents T.
3. Calculate the affine key values.
4. Apply the decryption formula.
5. Display the decrypted text.

Code:

```
#include <stdio.h>
#include <ctype.h>
int main() {
   char text[500];
   int i, c, a = 3, b = 15;
   printf("Enter ciphertext: ");
   fgets(text, sizeof(text), stdin);
   for (i = 0; text[i]; i++) {
      if (isalpha((unsigned char)text[i])) {
         c = toupper(text[i]) - 'A';
         text[i] = (9 * ((c - b + 26) % 26)) % 26 + 'A';
      }
   }
   printf("Plaintext: %s", text);
   return 0;
```

}

Input:

Enter ciphertext: B

Output:

